

Lesson 3

Date / /

(GAUSS' LAW)

1. Given that,
 $\vec{A} = (2\hat{i} + 3\hat{j}) \text{ m}^2$

$$\phi = ?$$

(a) $\vec{E} = 4\hat{i} \text{ N/C}$

We know that,

$$\begin{aligned}\phi &= \vec{E} \cdot \vec{A} \\ &= (4\hat{i}) \cdot (2\hat{i} + 3\hat{j}) \\ &= 8 \frac{\text{N-m}^2}{\text{C}}\end{aligned}$$

$$\therefore \phi = 8 \frac{\text{N-m}^2}{\text{C}}$$

(b) $\vec{E} = 5\hat{k} \text{ N/C}$

$$\begin{aligned}\therefore \phi &= \vec{E} \cdot \vec{A} \\ &= (5\hat{k}) \cdot (2\hat{i} + 3\hat{j}) \\ &= 0 \frac{\text{N-m}^2}{\text{C}}\end{aligned}$$

$$\therefore \phi = 0 \frac{\text{N-m}^2}{\text{C}}$$



Centre for Medical Research & Development (CMRD)

2. Given that,

$$q_{\text{enclosed}} = 1.8 \mu\text{C}$$
$$= 1.8 \times 10^{-6} \text{ C}$$

We know that,

$$\epsilon_0 = 8.86 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$$

According to Gauss Law,

$$\text{Electric flux } \phi = \oint \vec{E} \cdot d\vec{A}$$

$$= \frac{q_{\text{enclosed}}}{\epsilon_0}$$

$$\Rightarrow \phi = \frac{1.8 \times 10^{-6}}{8.86 \times 10^{-12}}$$

$$\phi = 2 \times 10^5 \frac{\text{N}\cdot\text{m}^2}{\text{C}}$$

$$\therefore \phi = 2 \times 10^5 \frac{\text{N}\cdot\text{m}^2}{\text{C}}$$

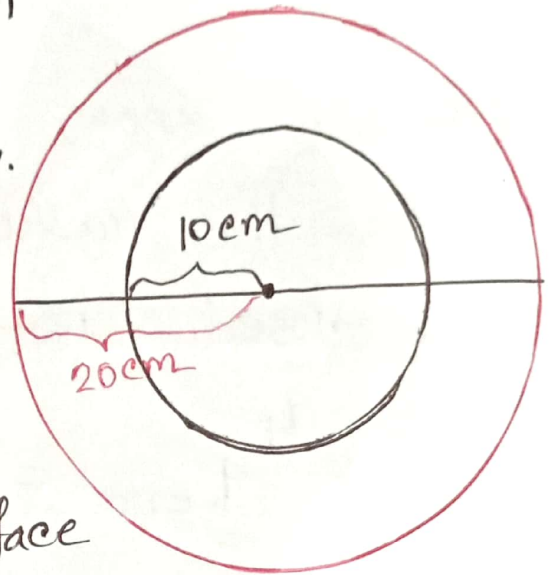
3. Given that,

$$\phi = -750 \frac{\text{N-m}^2}{\text{C}}$$

$$r = 10 \text{ cm}$$

(a) Electric flux depends on the net charge enclosed inside a body.

It does not depend on the size of the body. If the radius of the Gaussian surface is doubled, then the flux passing through the surface remains the same.



$$\therefore \phi = -750 \frac{\text{N-m}^2}{\text{C}}$$

$$(b) \phi = \frac{q}{\epsilon_0}$$

$$\therefore q = \phi \cdot \epsilon_0$$

$$= (-750) \cdot (8.85 \times 10^{-12}) \text{ C}$$

$$= -6.638 \times 10^{-9} \text{ C}$$

$$= -6.638 \text{ nC}$$

$$\therefore q = -6.638 \text{ nC}$$

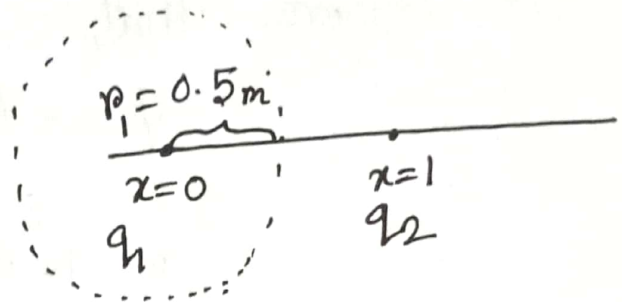


Centre for Medical Research & Development (CMRD)

4. Given that,

$$q_1 = 5 \text{ nC} \\ = 5 \times 10^{-9} \text{ C}$$

$$q_2 = -3 \text{ nC} = -3 \times 10^{-9} \text{ C}$$



(a) We know, $\phi = \frac{q_{\text{enc}}}{\epsilon_0}$

As the radius is 0.5 m, charge enclosed through this gaussian surface is q_1

$$\therefore q_{\text{enc}} = q_1$$

$$\therefore \phi_{0.5} = \frac{q_1}{\epsilon_0} = \frac{5 \times 10^{-9}}{8.85 \times 10^{-12}} \frac{\text{Nm}^2}{\text{C}}$$

$$\phi_{0.5} = 564.72 \frac{\text{Nm}^2}{\text{C}}$$

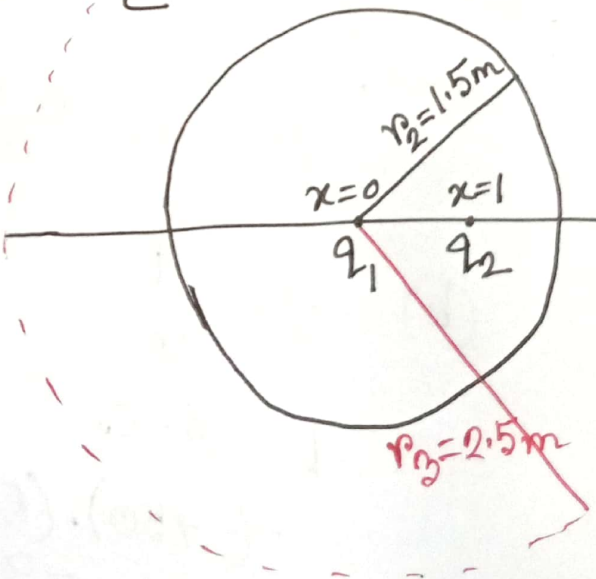
(b) $q_{\text{enc}} = q_1 + q_2$

$$= \{ (5 \times 10^{-9}) + (-3 \times 10^{-9}) \} \text{ C}$$

$$= 2 \times 10^{-9} \text{ C}$$

$$\phi_{1.5} = \frac{q_{\text{enc}}}{\epsilon_0}$$

$$= \frac{2 \times 10^{-9} \text{ C}}{8.85 \times 10^{-12} \frac{\text{C}^2}{\text{Nm}^2}}$$



$$\phi_{1.5} = 225.99 \frac{\text{Nm}^2}{\text{C}}$$

$$\phi_{1.5} \approx 226 \frac{\text{Nm}^2}{\text{C}}$$

$$(c) \quad r_3 = 2.5 \text{ m}$$

$$\begin{aligned} \therefore q_{\text{enc}} &= q_1 + q_2 \\ &= 2 \times 10^{-9} \text{ C} \end{aligned}$$

$$\phi_{2.5} = \frac{q_{\text{enc}}}{\epsilon_0}$$

$$\phi_{2.5} = \frac{2 \times 10^{-9}}{8.85 \times 10^{-12}} \frac{\text{Nm}^2}{\text{C}}$$

$$\phi_{2.5} \approx 226 \frac{\text{Nm}^2}{\text{C}}$$